Riccardo **Buscicchio** | Curriculum Vitæ

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Astrofisica relativistica, sviluppo di algoritmi di data analisi per inference Bayesiana e frequentista. Principali applicazioni: modellizzazione di missioni spaziali, rivelazione di segnali e stima dei parametri per astronomia ad onde gravitazionali, inferenza gerarchica di popolazione, ricerca di fondi stocastici.

Contatti

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Nazionalità: Italia

Sito web & e pubblicazioni: www.riccardobuscicchio.com – arXiv – ORCID

Incarichi accademici

Università degli Studi di Milano-Bicocca **Milano, Italia**
Assegnista di ricerca, Dipartimento di Fisica "G.Occhialini" *2021 - 2024*

- *Attività di ricerca*: sviluppo del segmento a terra per la data analisi di LISA per l' Agenzia Spaziale Italiana (Fase A).

Università degli Studi di Milano-Bicocca **Milano, Italia**
Assegnista di ricerca, Dipartimento di Fisica "G.Occhialini" *2024 - current*

- *Attività di ricerca*: sviluppo del segmento a terra per la data analisi di LISA per l' Agenzia Spaziale Italiana (Fase A).

Istruzione

University of Birmingham **Birmingham, Regno Unito**
Ph.D., School of Physics & Astronomy *2017-13/07/2022*

- *Supervisore*: A. Vecchio. La tesi ha prodotto 6 pubblicazioni a lista di autori corta.
- *Titolo della tesi*: Topics in Bayesian population inference for Gravitational Wave Astronomy

Questa tesi esplora diversi argomenti relativi all'inferenza bayesiana nell'astronomia a onde gravitazionali. Dall'inferenza gerarchica sulle popolazioni di di buchi neri binari di massa stellare, allo sviluppo di una routine completa di stima dei parametri per interferometri spaziali. Vengono inoltre affrontati i seguenti temi: popolazione di nane bianche binarie nelle galassie satellite della Via Lattea; vincoli dai fondi stocastici sul lensing di onde gravitazionali provenienti dalla fusione di binarie di stelle di neutroni e buchi neri; tecniche statistiche per l'inferenza congiunta su sorgenti multiple indistinguibili.

Università degli Studi di Pisa **Pisa, Italia**
Laurea Magistrale in Fisica Teorica *2013-2016*

- *Voto conseguito*: 110/110
- *Supervisore*: G. Cella. La tesi ha prodotto una pubblicazione a lista di autori corta.
- *Titolo della tesi*: An improved detector for non-gaussian stochastic background of gravitational waves.

Questa tesi ha esplorato l'idea di utilizzare il formalismo funzionale dei processi stocastici e della teoria dei campi classica per sviluppare un algoritmo migliorato per la rivelazione di fondi stocastici non gaussiani di onde gravitazionali.

Columbia University **New York NY, Stati Uniti**
Programma di scambio estivo INFN-NSF *Giu-Sett 2013*

- *Supervisori*: S. Marka, I. Bartos.

Abbiamo stimato il contributo al livello di rumore nei rivelatori di onde gravitazionali di seconda e terza generazione dovuto a sciami di raggi cosmici primari e secondari che attraversano gli specchi dell'interferometro.

Università degli Studi di Pisa **Pisa, Italia**
Laurea Triennale in Fisica *2008-2012*

- *Voto conseguito*: 109/110.
- *Titolo della tesi*: Banchi di template per la rivelazione delle onde gravitazionali: un'applicazione dell'Information Geometry.

Questa tesi ha esplorato l'idea di utilizzare il formalismo della geometria differenziale (come definito nel contesto della teoria dell'informazione) per sviluppare un algoritmo di posizionamento dei template nello spazio dei parametri delle sorgenti sfruttando la struttura di varietà non banale.

Indicatori bibliometrici

Pubblicazioni:

27 pubblicazioni short-author in riviste internazionali peer-reviewed
(di cui **7** articoli a primo autore e **5** di studenti supervisionati).
13 articoli di collaborazione con contributo sostanziale, pubblicati in riviste internazioni peer-reviewed
47 articoli di collaborazionane in totale, pubblicati in riviste internazioni peer-reviewed
6 articoli in fase pre-print,
2 altre pubblicazioni (tesi di dottorato, white papers, reviews)

Numero totale di citazioni: >14400. **h-index**: 23 (secondo record ADS e iNSPIRE).
Link a profili di citazione: ADS; iNSPIRE; arXiv; orcid.
Lista completa delle pubblicazioni disponibile a seguire e all'indirizzo
`www.riccardobuscicchio.com/publications`.
Lista completa dei seminari disponibile a seguire e all'indirizzo
`www.riccardobuscicchio.com/talks`.

Grant, Premi, & Riconoscimenti

Premi Accademici:

- Braccini PhD Thesis Prize, Menzione d'onore della Gravitational Wave International Committee. *2021*
- Michael Penston PhD Thesis Prize, Secondo premio della Royal Astronomical Society. *2021*

Grants:

- EuroHPC PRACE "LISAS-FIT" proposal, 100k CPUh su Leonardo BOOSTER *2023*
- Google Cloud for Researchers, 4kEUR Google Cloud Research Credits *2023*
- CINECA ISCRA Type-C project "LISA-MW" proposal, 10k CPUh presso il Centro Nazionale per High Performance Compu *2022*
- Grant di viaggio, Horizon 2020 AHEAD 2020 (High Energy Astrophysics) *2021*
- Grant di viaggio, American Physical Society, DGRAV Student Travel Grant *2020*
- Grant di viaggio, Institute of Physics Student Travel fund *2019*
- Grant di viaggio, Royal Astronomical Society, Regno Unito. *2018*

Supervisione, Tutoraggio, Insegnamento

Co-supervisione studenti di Dottorato:

- F. Nobili, Università dell'Insubria, *2024-2027*
- A. Spadaro, Università di Milano-Bicocca. *2022-2025*
- F. Pozzoli, Università dell'Insubria. *2022-2025*

Co-supervisione studenti di Laurea magistrale:

- L. Viganò, Università di Milano-Bicocca, Tesi magistrale *2024-2025*
- M. Bellotti, Università di Milano-Bicocca, Tesi magistrale. *2024-2025*
- D. Chirico, Università di Milano-Bicocca, Tesi magistrale. *2023-2024*
- S. Corbo, Politecnico di Milano, Tesi magistrale. *2023-2024*
- R. Rosso, Università di Pisa, Tesi magistrale. *2023-2024*
- G. Astorino, Università di Pisa, Tesi magistrale. *2023-2024*
- M. Piarulli, Università di Milano-Bicocca, Tesi magistrale. *2022-2023*

(ora studente di dottorato presso Univ. di Tolosa, Francia)

- A. Spadaro, Università di Milano-Bicocca, Tesi magistrale. *2021-2022*
  (ora studentessa di dottorato presso Università di Milano-Bicocca, Italia)
- A. Carzaniga, Università di Milano-Bicocca, Tesi magistrale. *2021-2022*
- A. Geminardi, Università di Milano-Bicocca, Tesi magistrale. *2021-2022*
  (ora studente di dottorato presso Univ. di Pavia, Italia)
- E. Finch, Università di Birmingham, Tesi magistrale. *2018*
- V. Spasova, Università di Birmingham, Tesi magistrale. *2018*

Co-supervisione studenti di Laurea Triennale:

- H. P. G. Carabajo, Università di Milano-Bicocca, Tesi triennale. *2023-2024*

Insegnamenti:

- Current and future challenges in GW astronomy, Corso di Dottorato, Milano-Bicocca (Italia). *2023*
- Mathematical physics and gravity (MAF900), Corso di dottorato, Univ. of Stavanger (Norvegia). *2023*
- Lezioni per il corso di Astrostatistica (F5802Q014/20), Laurea Magistrale in Astrofisica,Univ. di Milano-Bicocca. *2022*

Tutoring:

- Year 2 Python Computing Lab, Physics BSc, Birmingham. *2017-2021*
- Year 2 Maths for physicists, Physics BSc, Birmingham. *2017-2019*
- Year 4 Physics and communication skills, Physics BSc, Birmingham. *2019*

Academic service, editorial and research responsibilities

International collaboration responsibilities

Co-chair of the Coordination Unit L2D (ESA LISA Project Office, PA&QA) *2024-current*

Journal referee

- Physical Review Letters
- The Astrophysical Journal Letters
- Journal of Cosmology and Astroparticle Physics
- Monthly Notices of the Royal Astronomical Society
- Institute of Physics Trusted Reviewer Excellence program
- Physical Review D
- The Open Journal of Astrophysics
- NASA Technology Transfer Program
- Classical and Quantum Gravity

Editorial responsibilities

- Editorial board Lensing subgroup in the LIGO,Virgo, KAGRA collaboration *2023*
- Co-editor of a Living Review in relativity issue on "LISA data analysis" *2022-2023*

Conference organizer

- *LISA Distributed Data Processing Center June Workshop*, Milan, IT. *2025*
- IFPU focus week on "*Emerging methods in GW population inference*", Trieste, IT. *2024*
- *LISA Astrophysics Working Group Conference*, Birmingham, UK. *2022*
- *Gravitational-wave populations: what's next?*, Milan, Italy. *2023*
- *Gravitational-wave Excellence Alliance Training (GrEAT) PhD school*, Birmingham, UK. *2019*
- *Gravitational-wave Open Science Center First Open Data Workshop*, Remote *2019*

Outreach & public engagement

- Development of illustrations and animations for LISA Consortium *2023*
- Development of illustrations and graphic content for LIGO Magazine *2022-2023*
- Development of visualisation interface and skymaps content for GW alerts web and smartphone app. `https://chirp.sr.bham.ac.uk` *2022-2023*
- Organization of biweekly public engagement events *"Astronomy in the city"*, Birmingham, UK *2017-2021*
- Lectures to high-school students, Italy *since 2021*

Professional recognition and service

- Italian Habilitation to Associate Professorship in Astrophysics (ASN 02/C1). *2023*

- Secretary for the Council of Research Fellows, University of Milano-Bicocca *2024-2025*
- Research Fellow Representative, Department of Physics, University of Milano-Bicocca *2023-2025*
- French Qualification for teaching in Higher Education in Astrophysics (Sec.34). *2023*
  French Ministry of higher education and research (qualification no.23234388826).
- Meetings organizer "PhD meet and greet", University of Birmingham *2021*
- LSC Academic Advisory Committee. *2019-2021*

Memberships

- LISA Consortium, core member. *since 2018*
- Italian Center for Supercomputing (ICSC). *since 2021*
- TEONGRAV National Initiative (Gravity Theory) *since 2021*
  Italian National Institute for Nuclear Physics (INFN).
- LIGO, Virgo, Kagra Collaboration, full member. *since 2017*
- Italian Society of General Relativity and Gravitational Physics (SIGRAV) *since 2021*
- Istituto Nazionale di Astrofisica (INAF) *since 2021*
- American Physical Society (APS), member.
- Italian Physical Society (SIF), member *2021*
- Royal Astronomical Society (RAS), fellow. *2018-2021*

Competenze

Linguaggi di programmazione: Python (avanzato), Bash (avanzato), Julia (avanzato), Mathematica, Go, R (avanzato), Stan, C, Qt5.

Altri strumenti di programmazione: TensorFlow, LIGO lalsuite, LaTeX, controllo versione, strumenti HPC, containerization, continous integration, cloud computing, sviluppo web.

Lingue: Inglese (C2), Italiano (madrelingua), Francesse (A2)

Hobbies

Nuoto, corsa, arrampicata sportiva, fotografia. Fantascienza, musica elettronica.

Lista completa di pubblicazioni

Preprints a lista di autori breve e di collaborazione a cui ho contribuito significativamente.:

6. *Variability in the massive black hole binary candidate SDSS J2320+0024: no evidence for periodic modulation.*
F. Rigamonti, L. Bertassi, **R. Buscicchio**, F. Cocchiararo, S. Covino, M. Dotti, A Sesana, P. Severgnini.
arXiv:2505.22706[astro-ph.GA].
5. *Accelerating LISA inference with Gaussian processes.*
J. El Gammal, **R. Buscicchio**, G. Nardini, J. Torrado.
arXiv:2503.21871 [astro-ph.HE].
4. *Is your stochastic signal really detectable?.*
F. Pozzoli, J. Gair, **R. Buscicchio**, L. Speri.
arXiv:2412.10468 [astro-ph.IM].
3. *A test for LISA foreground Gaussianity and stationarity. I. Galactic white-dwarf binaries.*
R. Buscicchio, A. Klein, V. Korol, F. Di Renzo, C.J. Moore, D. Gerosa, A. Carzaniga.
arXiv:2410.08263 [astro-ph.HE].
2. *LISA Definition Study Report.*
M. Colpi, K. Danzmann, M. Hewitson, K. Holley-Bockelmann, et al. (incl. **R. Buscicchio**).
arXiv:2402.07571 [astro-ph.CO].
1. *The last three years: multiband gravitational-wave observations of stellar-mass binary black holes.*
A. Klein, G. Pratten, **R. Buscicchio**, P. Schmidt, C. J. Moore, E. Finch, A. Bonino, L. M. Thomas, N. Williams, D. Gerosa, S. McGee, M. Nicholl, A. Vecchio.
arXiv:2204.03423 [astro-ph.HE].

Papers in major peer-reviewed journals:

27. *Test for LISA foreground Gaussianity and stationarity: extreme mass-ratio inspirals.*
M. Piarulli, **R. Buscicchio**, F. Pozzoli, O. Burke, M. Bonetti, A. Sesana.
Physical Review D 111, (2025) 103047. arXiv:2410.08862 [astro-ph.HE].
26. *Cyclostationary signals in LISA: a practical application to Milky Way satellites.*
F. Pozzoli, **R. Buscicchio**, A. Klein, V. Korol, A. Sesana, F. Haardt.
Physical Review D 111, (2025) 063005. arXiv:2410.08274 [astro-ph.GA].
25. *Characterization of non-Gaussian stochastic signals with heavier-tailed likelihoods.*
N. Karnesis, A. Sasli, **R. Buscicchio**, N. Stergioulas.
Physical Review D 111, (2025) 022005. arXiv:2410.14354 [gr-qc].
24. *Stellar-mass black-hole binaries in LISA: characteristics and complementarity with current-generation interferometers.*
R. Buscicchio, J. Torrado, C. Caprini, G. Nardini, M. Pieroni, N. Karnesis, A. Sesana.
Journal of Cosmology and Astroparticle Physics 01 (2025) 084. arXiv:2410.18171 [astro-ph.HE].
23. *Stars or gas? Constraining the hardening processes of massive black-hole binaries with LISA.*
A. Spadaro, **R. Buscicchio**, D. Izquierdo–Villalba, D. Gerosa, A. Klein, G. Pratten.
Physical Review D 111, (2025) 023004. arXiv:2409.13011 [astro-ph.HE].
22. *Partial alignment between jets and megamasers: coherent or selective accretion?.*
M. Dotti, **R. Buscicchio**, F. Bollati, R. Decarli, W. Del Pozzo, A. Franchini.
Astronomy & Astrophysics 692 (2024) A233. arXiv:2403.18002 [astro-ph.GA].
21. *Expected insights on type Ia supernovae from LISA's gravitational wave observations.*
V. Korol, **R. Buscicchio**, Ruediger Pakmor, Javier Morán-Fraile, Christopher J. Moore, Selma E. de Mink.
Astronomy & Astrophysics 691 (2024) A44. arXiv:2407.03935 [astro-ph.HE].
20. *A weakly-parametric approach to stochastic background inference in LISA.*
F. Pozzoli, **R. Buscicchio**, C. J. Moore, A. Sesana, F. Haardt, A. Sesana.
Physical Review D 109, (2024) 083029. arXiv:2311.12111 [astro-ph.CO].
19. *A fast test for the identification and confirmation of massive black hole binary.*
M. Dotti, F. Rigamonti, S. Rinaldi, W. Del Pozzo, R. Decarli, **R. Buscicchio**.
Astronomy & Astrophysics 680 (2023) A69. arXiv:2310.06896 [astro-ph.HE].
18. *Glitch systematics on the observation of massive black-hole binaries with LISA.*
A. Spadaro, **R. Buscicchio**, D. Vetrugno, A. Klein, D. Gerosa, S. Vitale, R. Dolesi, W. J. Weber, M. Colpi.
Physical Review D 108 (2023) 123029. arXiv:2306.03923 [gr-qc].

17. *Implications of pulsar timing array observations for LISA detections of massive black hole binaries.*
N. Steinle, H. Middleton, C. J. Moore, S. Chen, A. Klein, G. Pratten, **R. Buscicchio**, E. Finch, A. Vecchio.
Monthly Notices of the Royal Astronomical Society 525 2 (2023). arXiv:2305.05955 [astro-ph.HE].

16. *Parameter estimation of binary black holes in the endpoint of the up-down instability.*
V. De Renzis, D. Gerosa, M. Mould, **R. Buscicchio**, L. Zanga.
Physical Review D 108 (2023) 024024. arXiv:2304.13063 [gr-qc].

15. *Improved detection statistics for non Gaussian gravitational wave stochastic backgrounds.*
M. Ballelli, **R. Buscicchio**, B. Patricelli, A. Ain, G. Cella.
Physical Review D 107 (2023) 124044. arXiv:2212.10038 [gr-qc].

14. *Detecting non-Gaussian gravitational wave backgrounds: a unified framework.*
R. Buscicchio, A. Ain, M. Ballelli, G. Cella, B. Patricelli.
Physical Review D 107 (2023) 063027. arXiv:2209.01400 [gr-qc].

13. *Detectability of a spatial correlation between stellar-mass black hole mergers and Active Galactic Nuclei in the Local Universe.*
N. Veronesi, E.M. Rossi, S. van Velzen, **R. Buscicchio**.
Monthly Notices of the Royal Astronomical Society 514 2 (2023). arXiv:2203.05907 [astro-ph.HE].

12. *Bayesian parameter estimation of stellar-mass black-hole binaries with LISA.*
R. Buscicchio, A. Klein, E. Roebber, C. J. Moore, D. Gerosa, E. Finch, A. Vecchio.
Physical Review D 104 (2021) 044065. arXiv:2106.05259 [astro-ph.HE].

11. *An Interactive Gravitational-Wave Detector Model for Museums and Fairs.*
S. J. Cooper, A. C. Green, H. R. Middleton, C. P. L. Berry, **R. Buscicchio**, E. Butler, C. J. Collins, C. Gettings, D. Hoyland, A. W. Jones, J. H. Lindon, I. Romero-Shaw, S. P. Stevenson, E. P. Takeva, S. Vinciguerra, A. Vecchio, C. M. Mow-Lowry, A. Freise.
American Journal of Physics 89 (2021) 702–712. arXiv:2004.03052 [physics.ed-ph].

10. *Evidence for hierarchical black hole mergers in the second LIGO–Virgo gravitational-wave catalog.*
C. Kimball, C. Talbot, C.P.L. Berry, M. Zevin, E. Thrane, V. Kalogera, **R. Buscicchio**, M. Carney, T. Dent, H. Middleton, E. Payne, J. Veitch, D. Williams .
Astrophysical Journal Letters 915 (2021) L35. arXiv:2011.05332 [astro-ph.HE].

9. *Testing general relativity with gravitational-wave catalogs: the insidious nature of waveform systematics.*
C. J. Moore, E. Finch, **R. Buscicchio**, D. Gerosa.
iScience 24 (2021) 102577. arXiv:2103.16486 [gr-qc].

8. *LoCuSS: The splashback radius of massive galaxy clusters and its dependence on cluster merger history.*
M. Bianconi, **R. Buscicchio**, G. P. Smith, S. L. McGee, C.P. Haines, A. Finoguenov, A. Babul.
Astrophysical Journal 911 (2021) 136. arXiv:2010.05920 [astro-ph.GA].

7. *Search for Black Hole Merger Families.*
D. Veske, A. G. Sullivan, Z. Marka, I. Bartos, K. R. Corley, J. Samsing, **R. Buscicchio**, S. Marka.
Astrophysical Journal Letters 907 (2021) L48. arXiv:2011.06591 [astro-ph.HE].

6. *Constraining the lensing of binary black holes from their stochastic background.*
R. Buscicchio, C. J. Moore, G. Pratten, P. Schmidt, M. Bianconi, A. Vecchio.
Physical Review Letters 125 (2020) 141102. arXiv:2006.04516 [astro-ph.CO].

5. *Constraining the lensing of binary neutron stars from their stochastic background.*
R. Buscicchio, C. J. Moore, G. Pratten, P. Schmidt, A. Vecchio.
Physical Review D 102 (2020) 081501 . arXiv:2008.12621 [astro-ph.HE].

4. *Measuring precession in asymmetric compact binaries.*
G. Pratten, P. Schmidt, **R. Buscicchio**, L. M. Thomas.
Physical Review Research 2 (2020) 043096. arXiv:2006.16153 [gr-qc].

3. *Populations of double white dwarfs in Milky Way satellites and their detectability with LISA.*
V. Korol, S. Toonen, A. Klein, V. Belokurov, F. Vincenzo, **R. Buscicchio**, D. Gerosa, C. J. Moore, E. Roebber, E. M. Rossi, A. Vecchio.
Astronomy & Astrophysics 638 (2020) A153. arXiv:2002.10462 [astro-ph.GA].

2. *Milky Way satellites shining bright in gravitational waves.*
E. Roebber, **R. Buscicchio**, A. Vecchio, C. J. Moore, A. Klein, V. Korol, S. Toonen, D. Gerosa, J. Goldstein, S. M. Gaebel, T. E. Woods.
Astrophysical Journal Letters 894 (2020) L15. arXiv:2002.10465 [astro-ph.GA].

1. *Label Switching Problem in Bayesian Analysis for Gravitational Wave Astronomy.*
R. Buscicchio, E. Roebber, J. M. Goldstein, C. J. Moore .
Physical Review D 100 (2019) 084041. arXiv:1907.11631 [astro-ph.IM].

Collaboration papers in major peer-reviewed journals, which I have substantially contributed to.:

13. *Search for gravitational-lensing signatures in the full third observing run of the LIGO-Virgo network.*
LIGO Scientific Collaboration, Virgo Collaboration, KAGRA collaboration.
Astrophysical Journal 970 (2021) 191. arXiv:2304.08393 [gr-qc].

12. *GWTC-2.1: Deep Extended Catalog of Compact Binary Coalescences Observed by LIGO and Virgo During the First Half of the Third Observing Run.*
LIGO Scientific Collaboration, Virgo Collaboration, KAGRA collaboration.
Physical Review D 109 (2024) 022001. arXiv:2108.01045 [gr-qc].

11. *The population of merging compact binaries inferred using gravitational waves through GWTC-3.*
LIGO Scientific Collaboration, Virgo Collaboration, KAGRA collaboration.
Physical Review X 13 (2021) 011048. arXiv:2111.03634 [astro-ph.HE].

10. *Tests of General Relativity with GWTC-3.*
LIGO Scientific Collaboration, Virgo Collaboration, KAGRA collaboration.
Physical Review D (accettato per la pubblicazione). arXiv:2112.06861 [gr-qc].

9. *Search for lensing signatures in the gravitational-wave observations from the first half of LIGO-Virgo's third observing run.*
LIGO Scientific Collaboration, Virgo Collaboration, KAGRA collaboration.
Astrophysical Journal Letters (2021) 923. arXiv:2105.06384 [gr-qc].

8. *GWTC-3: Compact Binary Coalescences Observed by LIGO and Virgo During the Second Part of the Third Observing Run.*
LIGO Scientific Collaboration, Virgo Collaboration, KAGRA collaboration.
Physical Review X 13 (2023) 041039. arXiv:2111.03606 [gr-qc].

7. *Observation of gravitational waves from two neutron star-black hole coalescences.*
LIGO Scientific Collaboration, Virgo Collaboration.
Astrophysical Journal Letters, 915, L5 (2021). arXiv:2106.15163 [astro-ph.HE].

6. *GWTC-2: Compact Binary Coalescences Observed by LIGO and Virgo During the First Half of the Third Observing Run.*
LIGO Scientific Collaboration, Virgo Collaboration.
Physical Review X 11 (2021) 021053. arXiv:2010.14527 [gr-qc].

5. *Population Properties of Compact Objects from the Second LIGO-Virgo Gravitational-Wave Transient Catalog.*
LIGO Scientific Collaboration, Virgo Collaboration.
Astrophysical Journal Letters 913 (2021) L7. arXiv:2010.14533 [astro-ph.HE].

4. *Upper Limits on the Isotropic Gravitational-Wave Background from Advanced LIGO's and Advanced Virgo's Third Observing Run.*
LIGO Scientific Collaboration, Virgo Collaboration, KAGRA collaboration.
Physical Review D 104 (2021) 022004. arXiv:2101.12130 [gr-qc].

3. *Binary Black Hole Population Properties Inferred from the First and Second Observing Runs of Advanced LIGO and Advanced Virgo .*
LIGO Scientific Collaboration, Virgo Collaboration.
Astrophysical Journal 882 (2019) L24. arXiv:1811.12940 [astro-ph.HE].

2. *Properties and astrophysical implications of the 150 Msun binary black hole merger GW190521.*
LIGO Scientific Collaboration, Virgo Collaboration.
Astrophysical Journal Letters 900 (2020) L13. arXiv:2009.01190 [astro-ph.HE].

1. *GW190521: A Binary Black Hole Merger with a Total Mass of 150 $M_\odot$.*
LIGO Scientific Collaboration, Virgo Collaboration.
Physical Review Letters 125 (2020) 101102. arXiv:2009.01075 [gr-qc].

Altre pubblicazioni, incluse tesi di dottorato, e altri articoli di collaborazione.:

2. *LISA - Laser Interferometer Space Antenna - Definition Study Report.*
The European Space Agency.
ESA-SCI-DIR-RP-002.

1. *Topics in Bayesian population inference for gravitational wave astronomy.*
R. Buscicchio.
PhD thesis.

Lista completa di seminari

Invited talks marked with *.

Seminari a conferenze nazionali e internazionali:

29.* *Emergence of Milky Way structure in the first year of LISA data.*
CERN UniGe Gravitational Wave meeting, Geneva, Switzerland, 2025/05/23.

28. *LISA stellar-mass black holes informed by the GWTC-3 population: event rates and parameters reconstruction.*
LISA Astrophysics Working Group Meeting 2024, Garching, Germany, 2024/11/05.

27.* *Astrophysics panel session.*
GRASP: Gravity Shape Pisa 2024, Pisa, Italy, 2024/10/24.

26.* *Beyond Gauss? A more accurate model for LISA astrophysical noise sources.*
Kavli Institute for Cosmology Seminars, Cambridge, United Kingdom, 2024/10/14.

25.* *Beyond Gauss? A more accurate model for LISA astrophysical noise sources.*
Heterogeneous Data and Large Representation Models in Science, Toulouse, France, 2024/10/01.

24. *LISA stellar-mass black holes informed by the GWTC-3 population: event rates and parameters reconstruction.*
15th International LISA Symposium, Dublin, Ireland, 2024/07/08.

23.* *LISA data analysis: from the stochastic background to the Milky Way.*
11th LISA Cosmology Working Group Workshop, Porto, Portugal, 2024/06/19.

22.* *An introduction to Bayesian Inference.*
International Pulsar Timing Array Student Week, Milan, Italy, 2024/06/17.

21.* *Statistical challenges in LISA data analysis.*
LAUTARO joint meeting, GSSI-University of Milano-Bicocca, Milano, Italy, 2024/04/17.

20. *From mHz to kHz: stochastic background implications on astrophysical sources and population reconstruction.*
LISA Astrophysics working group workshop, University of Milano-Bicocca, Milano, Italy, 2023/09/13.

19. *Non-gaussian gravitational wave backgrounds across the GW spectrum.*
XXV Sigrav conference on general relativity and gravitation, SISSA, Trieste, Italy, 2023/09/04.

18.* *LISA SGWB data analysis (session chair).*
Data Analysis Challenges for SGWB Workshop, CERN, Geneva, Switzerland, 2023/07/19.

17.* *Global Fit and foregrounds.*
LISA SGWB detection brainstorming, Univ. of Geneva, Geneva, Switzerland, 2023/07/17.

16.* *Beyond functional forms: non-parametric methods. (panelist talk).*
Gravitational-wave populations: What's next?, Milano-Bicocca, IT, 2023/07/01.

15. *The last three years : multiband gravitational-wave observations of stellar-mass binary black holes.*
LISA Astrophysics working group workshop, University of Birmingham, Birmingham, UK, 2022/06/23.

14. *The last three years : multiband gravitational-wave observations of stellar-mass binary black holes.*
American Physical Society (APS) April meeting, New York (NY), USA, 2022/04/12.

13. *Bayesian parameter estimation of stellar-mass black-hole binaries with LISA.*
XXIV Sigrav conference on general relativity and gravitation, Urbino, Italy, 2021/09/08.

12. *Chirp: a web and smartphone application for visualization of gravitational-wave alerts.*
(remote) 14th Amaldi Conference on Gravitational Waves, 2021/07/21.

11.* *Search for lensing signatures in the gravitational-wave observations from the first half of LIGO-Virgo's third observing run.*
2nd EPS conference on gravitation, (remote, on behalf of LVK), 2021/05/27.

10.* *Bayesian parameter estimation of stellar-mass black-hole binaries with LISA.*
LISA Data Challenge meeting, (remote), 2021/06/17.

9.* *Search for lensing signatures in the gravitational-wave observations from the first half of LIGO-Virgo's third observing run.*
Webinar on behalf of the LVK collaboration, (remote), 2021/05/27.

8. *Milky Way Satellites Shining Bright in Gravitational Waves.*
13th LISA Symposium, (remote), 2020/09/13.

7. *Constraining the Lensing of Binary Black Holes from Their Stochastic Background.*
LISA Sprint workshop, Center for Computational Astrophysics, Flatiron Institute, New York (NY), USA, 2020/03/04.

6. *Multiple source detection in GW astronomy: the label switching problem.*
30th Texas Symposium, University of Portsmouth, Portsmouth, UK, 2019/12/12.

5. *Non-gaussian Stochastic background search with importance sampling.*
LIGO, Virgo, KAGRA September meeting, Warsaw, Poland, 2019/09/01.

4. *Hierarchical nonparametric density estimation for population inference.*
LIGO, Virgo, KAGRA March meeting, Winsconsin, USA, 2019/03/18.
3. *An improved detector for non-Gaussian stochastic background.*
Stochastic Background Data Analysis for LISA meeting, Instituto de Fisica Teorica, Madrid, Spain, 2019/–/–.
2. *Fast Evaluation of Campbell processes N–point correlation functions.*
Astro Hack Week: Data Science for Next-Generation Astronomy, Lorentz Center, Leiden, The Netherlands, 2018/08/01.
1. *Stochastic Gravitational Wave Background Data Analysis for Radler.*
5th LISA Cosmology Working Group workshop, Physicum, University of Helsinki, Helsinki, Finland, 2018/06/01.

Seminari presso istituti di ricerca e università:

10.* *Fast LISA inference using Gaussian processes.*
University of Geneva, Geneva, Switzerland, 2025/05/21.
9.* *Emergence of Milky Way structure in the first year of LISA data.*
Department of Physics, Pisa, Italy, 2025/05/16.
8.* *Statistical challenges in GW inference: an application of field theory to direct population reconstruction in LISA.*
APP seminar, SISSA, Trieste, Italy, 2024/05/06.
7.* *GRAF: Gravitational waves data and global fit.*
Department of Physics, University of Milano-Bicocca, Milan, Italy, 2023/12/14.
6.* *LISA global inference: statistical and modelling challenges for the Milky Way.*
Max Planck Institute for Astrophysics, Munchen, Germany, 2023/11/29.
5.* *LISA Global inference: modelling, statistical, and computational challenges.*
Department of Physics, University of Pisa, Pisa, Italy, 2023/10/04.
4.* *Gravitational waves in the many sources, many detectors era.*
Institute for Mathematics and Physics, University of Stavanger, Stavanger, Norway, 2022/09/29.
3.* *Stellar mass binary black holes : what, when, and where.*
Astroparticule et cosmologie, Universitè Paris Citè, Paris, France, 2022/06/12, (remote).
2.* *The last three years: multiband gravitational-wave observations of stellar-mass binary black holes.*
Physics Department, Columbia University, New York (NY), USA, 2022/04/07.
1.* *Set the alarm : Bayesian parameter estimation of stellar-mass black-hole binaries with LISA.*
Sun Yat-sen University, Zhuhai, China, 2021/07/30, (remote).